

To Shrink or to Select? Evaluating Finite-Sample Performance in Set-Identified Structural VARs

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Abstract

This paper evaluates specification strategies for estimating structural Impulse Response Functions (IRFs) in Vector Autoregressions (VARs), updating traditional lag-selection evaluations with modern Bayesian methods and set identification. Using Monte Carlo simulations calibrated to the global oil market, we compare information criteria (AIC, BIC, HQC) against Bayesian VARs (BVAR) in recovering structural shocks identified via sign restrictions. Results confirm that AIC outperforms stricter criteria (BIC, HQC) in finite samples by minimizing bias. However, BVAR models generally dominate frequentist lag selection, especially for high-order models, by leveraging prior shrinkage rather than strictly truncating lags. Furthermore, optimizing the shrinkage hyperparameter via Marginal Data Density (MDD) imposes excessive shrinkage, yielding suboptimal structural estimates. This highlights a critical divergence between forecasting and structural inference, concluding that a moderately loose BVAR prior offers a robust approach for accurate IRF recovery.

Keywords: Structural Vector Autoregression (SVAR), Set Identification, Bayesian VAR (BVAR), Lag Order Selection, Monte Carlo Simulation

JEL Codes: C11, C15, C32, C52

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This paper has been completed in conjunction with (CITATION). The main difference is that this paper extends the method space by introducing BVAR, whereas (CITATION) implements BMA. Therefore, both the last subsection of the theoretical introduction, Subsection 2.4, as well as Section 4 and Section 6 differ. Replication files are available in the [GitHub repository](#) for both papers.

1 Introduction

Structural Vector Autoregressions (VARs) remain the cornerstone of a large part of empirical macroeconomics, especially for monetary policy, shock transmission, and financial markets. While forecasting may be the main goal for some researchers, many economists also use it to estimate the relationships among factors that determine macroeconomic movements. For this purpose, Impulse Response Functions (IRFs) are derived from the VAR parameters, establishing how a shock in one variable affects the development of another period, both contemporaneously and over horizons of several months or even years. Using this transformation, structural VARs are continually adapted to answer complex questions, such as how speculative trading and real-time news shocks drive global oil market dynamics (Kilian and Murphy, 2014; Gazzani et al., 2024; Castelnovo et al., 2024), how high-frequency monetary policy surprises transmit through corporate credit spreads (Gertler and Karadi, 2015; Ferreira et al., 2023), and how these policy shifts impact global commodity prices (Miranda-Pinto, 2023). The specification of the VAR model, as well as the method to decompose it into its structural form, are the two crucial specifications that a researcher has to decide on. Traditionally, VAR models were specified with a lag order determined by economic theory or an information criterion (IC) and then estimated by Ordinary Least Squares (OLS) or Maximum Likelihood (ML). The past 15 years have seen a surge in the popularity of Bayesian methods, in which not a single lag order is specified, but lag orders are weighted and/or shrunk based on a prior and a posterior probability distribution (Blake and Mumtaz, 2015). These methods enable the researcher not only to rely on a single specification but also to incorporate a larger model or set of models while still keeping variance low. The methodology for estimating IRFs has also advanced significantly since Sims (1980) showed that structural IRFs can be retrieved via Cholesky decomposition, i.e., a predetermined causal ordering of the relationships between variables. Newer methods typically require weaker assumptions and adopt a more agnostic approach to inferring economic relationships. The dominant approach nowadays is set identification using sign restrictions informed by economic theory (Uhlig, 2005; Rubio-Ramírez et al., 2010). Unlike in Cholesky decomposition, all variables can simultaneously react to each other’s shocks, and sign restrictions can reduce the set of possible IRFs to those that fit the researcher’s assumptions. This yields a set of admissible IRFs. Typically, the median target—the specific IRF closest to the point-wise median of the admissible set—is selected as the primary result (Fry and Pagan, 2011).

This study draws inspiration from Ivanov and Kilian (2005), the first paper to examine which VAR specification best retrieves the true IRF using Monte Carlo simulations. In 2005, this meant testing criteria for lag selection to assess how well they fit VAR models and retrieve structural IRFs via Cholesky decomposition. We aim to update this analysis to the more recent world of VAR methodology, where IRFs are identified via

sign restrictions and estimation relies heavily on Bayesian methods rather than solely on information criteria.. We construct a realistic DGP based on Kilian and Murphy (2014)’s study of the oil market, using the set identification method. Then we compare different methods across repeated simulations by how well they can retrieve the true IRF, using the median target of their identified set. This study compares the information criteria AIC, BIC, and HQC with BVAR estimates using different overall shrinkage hyperparameters λ_1 . We find that BVARs are a well-performing alternative to OLS-estimated VAR models with IC-selected lag orders, beating the best criterion, AIC, in most scenarios. We could not deduce a clear pattern for which λ_1 -values shall be optimally employed depending on sample size T and true lag order p_0 . We attribute this to the fact that both higher and lower shrinkage can have the same overall effect: one reduces variance, the other reduces bias. Additionally, we suspect that our simulation still contains too much sampling variance, as computational restrictions did not allow for sufficient iterations. Optimizing λ_1 via the marginal data density (MDD), a metric geared toward out-of-sample forecasting, generally favors tighter λ_1 -values in order to shrink strongly. Taking the results for different λ_1 specifications into account, this is found to be suboptimal, although still an improvement over AIC in most cases. We hence show that the goals of forecasting and IRF analysis do not necessarily work together in VAR estimation.

The rest of the paper is structured as follows: Section 2 introduces the reader to the theoretical foundations needed to understand the methodology of this simulation study. It first introduces the VAR framework, Section 2.1, then explains the sign restriction approach for IRF decomposition, Section 2.2, and explains information criteria as well as the BVAR method, Section 2.3 and 2.4. We aim to both convey the intuition behind these methods as well as derive key properties. Section 3 explains the simulation design and Section 4 presents our results. Finally, the conclusion, Section 6, summarizes and discusses our key findings.

2 Theoretical Background

2.1 VAR Framework and IRFs

The reduced form of a vector autoregression, $\text{VAR}(p_0)$, establishes a relationship between current observations of a set of variables with the past observations of these variables up until a pre-determined lag p_0 :

$$Y_t = v + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_{p_0} Y_{t-p_0} + e_t \quad (1)$$

Here, Y_t is a $(K \times 1)$ vector of endogenous variables, v is a vector of intercepts, $A_1, \dots, A_{p_0}$ are $(K \times K)$ coefficient matrices, e_t is a white-noise innovation vector with $\mathbb{E}[e_t] = 0$ and covariance matrix $\mathbb{E}[e_t e_t'] = \Sigma$, and p_0 denotes the lag order. This is the

standard reduced-form representation of a VAR process (Lütkepohl, 2005; Sims, 1980). However, reduced-form innovations are composed of structural shocks and are therefore uninformative about the effects of any individual shock. To identify the structural shocks of interest, restrictive assumptions must be imposed. The relationship between the two is given by:

$$e_t = B\epsilon_t \quad (2)$$

where ϵ_t denotes the vector of mutually uncorrelated structural shocks (that are normalized to unit variance, i.e., $\mathbb{E}[\epsilon_t \epsilon_t'] = I$) and B is a matrix satisfying $\Sigma = BB'$. Since there are infinitely many candidate matrices B (for any valid factor $\tilde{B}$ and any orthogonal matrix D , the product $\tilde{B}D$ also satisfies $\Sigma = (\tilde{B}D)(\tilde{B}D)'$), restrictive assumptions are required to single out a unique B . The literature offers several strategies for this purpose, including short-run (recursive) restrictions (Sims, 1980; Christiano et al., 1999), long-run restrictions (Blanchard and Quah, 1989), and sign restrictions (Faust, 1998; Uhlig, 2005; Rubio-Ramírez et al., 2010); see Kilian and Lütkepohl (2017) for a comprehensive treatment. Following Ivanov and Kilian (2005), we do not commit to a single scheme mentioned above but instead adopt, for each data-generating process, the identifying assumptions used in the original study on which it is based.

In general, interpreting the dynamics implied by VAR coefficients is difficult. Therefore, practitioners use Impulse Response Functions (IRF) to summarize the main dynamics that they want to explain. To obtain the coefficient set that constructs the impulse response function, we express the VAR model in terms of an infinite VMA process. Using (1):

$$Y_t = [I - A_1L - A_2L^2 - \dots - A_{p_0}L^{p_0}]^{-1}v + [I - A_1L - A_2L^2 - \dots - A_{p_0}L^{p_0}]^{-1}e_t \quad (3)$$

Which is equivalent to:

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i e_{t-i} \quad \text{and} \quad \Phi_0 = I_K \quad (4)$$

where $\mu = [I - A_1 - \dots - A_{p_0}]^{-1}v$ is the unconditional mean vector and the moving average coefficient matrices are obtained recursively as $\Phi_i = \sum_{j=1}^i A_j \Phi_{i-j}$ where $A_j = 0$ for $j > p_0$ (see appendix for detailed derivation). Finally, in order to obtain the structural impulse response functions, we plug the expression given in (2) into (4):

$$Y_t = \mu + \sum_{i=0}^{\infty} \Phi_i B \epsilon_{t-i} = \mu + \sum_{i=0}^{\infty} \Theta_i \epsilon_{t-i} \quad (5)$$

Here, the matrices Θ_i are the structural impulse response matrices, and they contain

the coefficients required to construct the impulse response function. To illustrate, the equation for the first endogenous variable can be written as:

$$Y_{1t} = \mu_1 + \sum_{k=1}^K \theta_{1k,0} \epsilon_{kt} + \sum_{k=1}^K \theta_{1k,1} \epsilon_{kt-1} + \sum_{k=1}^K \theta_{1k,2} \epsilon_{kt-2} + \dots \quad (6)$$

where $\theta_{jk,i}$ denotes the response of variable j to structural shock k at horizon i , i.e., the (j,k) element of Θ_i .

The structural impulse responses in (6) depend on the model through two distinct channels. First, the moving average coefficients Φ_i are determined by the reduced-form autoregressive coefficients $A_1, \dots, A_{p_0}$, so the impulse responses depend on the lag order p_0 ; in practice, p_0 is unknown and must be estimated, which is the subject of the lag-order selection criteria discussed in Section 3. Second, the mapping from reduced-form to structural form impulse responses requires the identification of B introduced in (2). Since there are infinitely many candidate matrices B (for any valid factor $\tilde{B}$ and any orthogonal matrix D , the product $\tilde{B}D$ also satisfies $\Sigma = (\tilde{B}D)(\tilde{B}D)'$), restrictive assumptions are required to single out a unique B . The literature offers several strategies for this purpose, including short-run (recursive) restrictions, often computed by Cholesky decomposition (Sims, 1980; Christiano et al., 1999), long-run restrictions (Blanchard and Quah, 1989), and sign restrictions (Faust, 1998; Canova and Nicoló, 2002; Uhlig, 2005; Rubio-Ramírez et al., 2010); see Kilian and Lütkepohl (2017) for a comprehensive treatment. We calibrate our data-generating process to the model of Kilian and Murphy (2014) and accordingly employ a sign-restriction algorithm to identify the impact matrix (though we omit the elasticity bounds for computational feasibility). A key feature of this approach is that sign restrictions deliver only partial identification: rather than a single impact matrix, they identify a *set* of candidate matrices consistent with the a priori restrictions implied by economic theory. This raises several questions about the identification strategy, which are the focus of the next subsection.

Structural Vector Autoregressions (VARs) remain the cornerstone of a large part of empirical macroeconomics, especially for monetary policy, shock transmission, and financial markets. While forecasting may be the main goal for some actors, especially central banks or national economic institutes, many macroeconomists also use it to estimate the relationships among factors that determine macroeconomic movements. For this purpose, Impulse Response Functions (IRFs) are derived from the VAR parameters, establishing how a shock in one variable affects the development of another period, both contemporaneously and over horizons of several months or even years. For example, it may be applied to study how shocks in speculative oil demand affect oil demand and supply flows (Kilian and Murphy, 2014) or how interest rate shocks impact credit costs (Gertler and Karadi, 2015) (change to add newer papers citations). The specification of the VAR model, as well as the method to decompose it into its structural form, are the

two crucial specifications that a researcher has to decide on. Traditionally, VAR models were specified with a lag order determined by economic theory or an information criterion (IC) and then estimated by Ordinary Least Squares (OLS) or Maximum Likelihood (ML). The past 15 years have seen a surge in the popularity of Bayesian methods, in which not a single lag order is specified, but lag orders are weighted and/or shrunk based on a prior and a posterior probability distribution (CITATION). These methods enable the researcher not only to rely on a single specification but also to incorporate a larger model or set of models while still keeping variance low. The methodology for estimating IRFs has also advanced significantly since Sims (1980) showed that structural IRFs can be retrieved via Cholesky decomposition, i.e., a predetermined causal ordering of the relationships between variables. Newer methods typically require weaker assumptions and adopt a more agnostic approach to inferring economic relationships. The dominant approach nowadays is set identification using sign restrictions informed by economic theory (Uhlig, 2005; Rubio-Ramírez et al., 2010). Unlike in Cholesky decomposition, all variables can simultaneously react to each other’s shocks, and sign restrictions can reduce the set of possible IRFs to those that fit the researcher’s assumptions. In the end, there is a set of admissible IRFs of which the median target, the IRF overall closest to the median of all IRFs, is usually selected as the main result (Fry and Pagan, 2011).

This study draws inspiration from Ivanov and Kilian (2005), the first paper to examine which VAR specification best retrieves the true IRF using Monte Carlo simulations. In 2005, this meant testing criteria for lag selection to assess how well they fit VAR models and retrieve structural IRFs via Cholesky decomposition. We aim to update this analysis to the more recent world of VAR methodology, where IRFs are determined by set identification and estimation relies not only on information criteria but also on Bayesian methods. We construct a realistic DGP based on Kilian and Murphy (2014)’s study of the oil market, using the set identification method. Then we compare different methods across repeated simulations by how well they can retrieve the true IRF, using the median target of their identified set. This study compares the information criteria AIC, BIC, and HQC with BVAR estimates using different overall shrinkage hyperparameters λ_1 . We find that BVARs are a well-performing alternative to OLS-estimated VAR models with IC-selected lag orders, beating the best criterion, AIC, in most scenarios. We could now deduce a clear pattern for which λ_1 -values shall be optimally employed depending on sample size T and true lag order p_0 . We attribute this to the fact that both higher and lower shrinkage can have the same overall effect: one reduces variance, the other reduces bias. Additionally, we suspect that our simulation still contains too much sampling variance, as computational restrictions did not allow for sufficient iterations. Optimization of λ_1 using marginal density distribution (MDD), which optimizes on out-of-sample performance, is found not to generally favor lower values of λ_1 , which is found to be suboptimal for the purpose of IRF decomposition. We therefore show that the goals of forecasting and IRF

analysis do not necessarily work together in VAR estimation.

The rest of the paper is structured as follows: Section 2 introduces the reader to the theoretical foundations needed to understand the methodology of this simulation study. It first introduces the VAR framework, Section 2.1, then explains the sign restriction approach for IRF decomposition, Section 2.2, and explains information criteria as well as the BVAR method, Section 2.3 and 2.4. We aim to both convey the intuition behind these methods as well as derive key properties. Section 3 explains the simulation design and Section 4 presents our results. Finally, the conclusion, Section 6, summarizes and discusses our key findings.

2.2 Sign Restrictions

Point identification strategies, such as the Cholesky decomposition, impose an assumption restrictive enough to single out a unique impact matrix; in the Cholesky case, this is a recursive ordering that determines the contemporaneous causal structure among the endogenous variables. Uhlig (2005), however, criticizes such strategies on the grounds that these restrictive assumptions may not reflect the underlying economic reality, but instead may impose the very conclusions the researcher expects to find.

Uhlig (2005) proposes an alternative he terms an "agnostic" identification procedure. The agnostic approach imposes only minimal restrictions—and, crucially, leaves the response of the variable of central interest entirely unrestricted—so that the answer to the research question is determined by the data rather than by the identifying assumptions. The restrictions that are imposed are deliberately weak: they consist only of sign restrictions on a few responses that can be reasonably agreed upon across economists. This minimalism comes at a cost, however. Imposing only sign restrictions does not single out a unique impact matrix; instead, it identifies a set of candidate impact matrices consistent with the assigned signs. This set-identification (or partial identification) strategy is less restrictive, but it comes at the cost of ambiguity, since the object of interest is no longer a single value but a range. Moreover, as emphasized by Uhlig (2005), sign restrictions introduce a further choice that point-identifying schemes do not: the horizon over which the restrictions are imposed. In addition to specifying the sign of a response on impact, the practitioner must decide how many subsequent periods that sign is required to hold—the longer this horizon, the more candidate models are discarded and the narrower the identified set becomes.

This trade-off is described by Manski (2003) as the Law of Decreasing Credibility: the credibility of inference decreases as the strength of the maintained assumptions increases. Point identification strategies impose stronger assumptions and are therefore more informative—they reduce the identified object to a single point—but they are less credible, since their validity rests entirely on assumptions that may not hold. Partial

identification strategies, by contrast, are less informative—they deliver a trimmed range rather than a single element—but more credible, precisely because they rely on weaker and more defensible assumptions.

Here, we follow Kilian and Murphy (2014) with slight differences due to computational feasibility.¹ First, we fit VAR(p_0) model to the data of Kilian and Murphy (2014) and obtain the reduced-form autoregressive coefficients ($\hat{A}$) and the variance-covariance matrix of the reduced-form errors ($\hat{\Sigma}$). We then factor this covariance matrix using Cholesky decomposition ($\hat{\Sigma} = PP'$).² Next, following Rubio-Ramírez et al. (2010), we generate random orthogonal matrices: we draw a matrix of independent standard normal entries, compute its QR decomposition, and normalize the sign of each column so that the diagonal of (R) is positive. This procedure yields orthogonal matrices (Q) (with $QQ' = I_K$) that are uniformly distributed with respect to the Haar measure, ensuring that the space of admissible rotations is sampled without bias. For each draw, we form a candidate impact matrix ($\tilde{B} = PQ$) and check whether it satisfies the imposed sign restrictions; candidates that violate the restrictions are discarded. Finally, we retain a select number of admissible impulse response structures and select the median target as our true IRF (population response). For the sake of clarity, we find it appropriate to outline the step-by-step blueprint of this algorithm:

1. Draw a ($K \times K$) matrix W , where each element is drawn independently from a standard normal distribution: $W_{ij} \sim N(0, 1)$.
2. Compute the QR decomposition $W = QR$, where Q is an orthogonal matrix (the rotation matrix) and R is upper triangular. Normalize the sign of each column of Q so that the diagonal of R is positive, which renders the decomposition unique and ensures that Q is distributed according to the Haar measure.
3. Construct the candidate impact matrix $\tilde{B} = PQ$, and then check whether $\tilde{B}$ satisfies the imposed restrictions.
4. Repeat this process by drawing new W matrices (i.e., obtaining new rotation matrices Q) until the desired number of admissible models is reached.
5. Among the retained models, select the median target m^* as the representative (true) model.

¹Kilian and Murphy (2014) argue that agnostic sign restrictions alone are not sufficient to narrow the set of admissible models adequately, and they therefore impose additional quantitative restrictions; bounds on the implied oil supply and demand elasticities. Additionally, they set dynamic restrictions for a horizon of 12 months. We omit the elasticity restrictions and reduce the dynamic restrictions to 3 months for reasons of computational feasibility: in our finite samples, the acceptance rate under the elasticity bounds is prohibitively low, so that obtaining a sufficient number of admissible models in each of the many Monte Carlo iterations would require a computationally unachievable number of draws.

²Kilian and Murphy (2014) uses eigendecomposition here. However, it doesn't make any difference. Any two factors satisfying $BB' = \tilde{B}\tilde{B}' = \hat{\Sigma}$ differ only by an orthogonal rotation ($B = \tilde{B}D$). Since our procedure applies the full set of Haar-distributed rotations to the initial factor, the resulting set of admissible models is invariant to the initial decomposition. The only element we need is an initial factor to run the algorithm.

The selection of a representative model is a much-debated issue in the literature, particularly the comparison between the pointwise median and the median target. Since we must obtain a single population response rather than a set—otherwise the mean squared error (MSE) could not be computed against a unique true IRF—we adopt the approach most appropriate for our design. Fry and Pagan (2011) criticize the pointwise median for a compelling reason: the pointwise median does not correspond to any single admissible model, as it combines responses drawn from different models with different impact matrices. Consequently, this synthetic profile may appear plausible but need not correspond to any B satisfying $\hat{\Sigma} = BB'$. This implies that the shocks may not be properly isolated, so the pointwise median cannot be used for variance decomposition. To address this, Fry and Pagan propose the median target approach:

$$m^* = \arg \min_m \sum_{j,k,i} \left(\frac{\theta_{jk,i}^{(m)} - \text{med}_{jk,i}}{\sigma_{jk,i}} \right)^2, \quad (7)$$

where $\text{med}_{jk,i} = \text{median}_m \{\theta_{jk,i}^{(m)}\}$, $\sigma_{jk,i} = \text{std}_m \{\theta_{jk,i}^{(m)}\}$.

Solving this minimization yields the admissible model whose impulse responses are closest to the pointwise median ($m^* \in M$). Since m^* is an actual admissible model rather than a synthetic construct, it satisfies $\hat{\Sigma} = BB'$ (which makes sure that shocks are orthogonalized) and can be used for variance decomposition. We adopt this approach due to the coherence concerns described above and take the median target as our representative true IRF whereby its specific impact matrix $\tilde{B}$ determines the DGP.³

Identification is an essential part of recovering the underlying impulse response function. As noted earlier, however, it is not the only factor that matters: the specification of the reduced-form VAR also plays a role. Recall that $\Phi_i = \sum_{j=1}^i A_j \Phi_{i-j}$ where $A_j = 0$ for $j > p_0$, so the moving-average coefficients, and hence the impulse responses, depend on the lag order (p_0). The practitioner must therefore choose the lag order carefully. We address this in the next section.

2.3 Information Criteria

The true lag order p_0 is a priori unknown when estimating a reduced-form VAR; it is a matter of specification. Adding too many lags may lead to overfitting, which inflates the variance of the impulse response estimates by requiring the estimation of additional parameters. Underfitting is also problematic, as it introduces bias. The bias-variance trade-off is therefore central to lag-order selection. Traditionally, information criteria are used to address this issue, and their working principle is essentially the same (Lütkepohl,

³For a visualization of the IRFs selected for the DGP, see Figure 4 in the Appendix.

2005, chap. 4):

$$\text{IC}(p) = \ln |\hat{\Sigma}(p)| + \underbrace{c_T, \frac{K^2 p}{N}}_{\text{penalty term}} \quad (8)$$

Information criteria are formulas that embody the bias-variance trade-off described above. The $\ln |\hat{\Sigma}(p)|$ term is the log-determinant of the reduced-form error covariance matrix, estimated by quasi-maximum likelihood; this term penalizes underfitting, since a model with too few lags leaves $\hat{\Sigma}(p)$ large and thus raises $\text{IC}(p)$. The Penalty weight c_T depends on the specific criterion, and the penalty term increases with the lag order; the second component therefore penalizes overfitting. The lag order is estimated by:

$$\hat{p} = \arg \min p \text{IC}(p), \quad 1 \leq p \leq \bar{p} \quad (9)$$

Following Ivanov and Kilian (2005), we set the maximum lag order to $\bar{p} = 12$ for our monthly data, corresponding to one year. This choice exceeds the largest true lag order considered $p_0 = 10$, ensuring that all criteria can both identify and overfit relative to the true order, and it makes our results directly comparable to those of Ivanov and Kilian (2005).⁴

Although there are different types of $\text{IC}(p)$, three are mainly used:

$$\text{AIC}(p) = \ln |\hat{\Sigma}(p)| + \frac{2}{N}, K^2 p \quad (10)$$

The Akaike Information Criterion (AIC) (Akaike, 1974) is the least overfitting-averse of these three criteria. It uses $c_T = 2$ and tends to overfit in practice.⁵

$$\text{BIC}(p) = \ln |\hat{\Sigma}(p)| + \frac{\ln N}{N}, K^2 p \quad (11)$$

The Bayesian Information Criterion (BIC), also called the Schwarz Information Criterion (Schwarz, 1978), penalizes overfitting the most. It uses $c_T = \ln N$ and is a consistent estimator of the true lag order $\left(\text{plim}_{N \rightarrow \infty} \hat{p}^{\text{BIC}} = p_0 \right)$.

$$\text{HQC}(p) = \ln |\hat{\Sigma}(p)| + \frac{2 \ln \ln N}{N}, K^2 p \quad (12)$$

The Hannan-Quinn Information Criterion (Hannan and Quinn, 1979) is neither as harsh

⁴We only include AIC, BIC and HQC as information criteria and omit the Likelihood Ratio tests and the Lagrange Multiplier test for lag selection as these have become practically irrelevant in the last 20 years for macroeconomic time series analysis and were also shown to perform poorly in Ivanov and Kilian (2005)

⁵In the literature, the AIC is generally regarded as prone to overfitting. In the VAR setting, however, this tendency is attenuated: adding one more lag requires estimating an entire $(K \times K)$ coefficient matrix, so the penalty grows by $\frac{2K^2}{N}$ at each step. Because each additional lag is this costly, even the AIC—the most lenient of the three criteria—can underfit in practice.

against overfitting as AIC nor as tolerant as BIC. It uses $c_T = 2 \ln \ln N$ as its penalty weight and, like BIC, is a consistent estimator of the true lag order.

The three criteria share the same fit term and differ only in the penalty weight c_T , which equals 2 for the AIC, $2 \ln \ln N$ for the HQC, and $\ln N$ for the BIC. A heavier penalty makes each additional lag more costly and therefore yields a more parsimonious model. Comparing the weights gives:

$$2; \leq; 2 \ln \ln N; \leq; \ln N \quad (13)$$

where the first inequality holds for $N \geq 16$ and the second for all N .⁶ Consequently, the implied lag-order estimates satisfy:

$$\hat{p}^{\text{BIC}}; \leq; \hat{p}^{\text{HQC}}; \leq; \hat{p}^{\text{AIC}} \quad (14)$$

so that the BIC is the most parsimonious criterion and the AIC the least, with the HQC in between (Ivanov and Kilian, 2005). Since $N \geq 16$ holds across all sample sizes in our study, this ordering applies throughout. As shown in Table 1, our Monte Carlo Simulation confirms this ordering. The penalization of BIC is evidenced to be so strong that the lag order selected, $\hat{p}^{\text{BIC}}$, always stays close to 2, regardless of the sample size T and the true lag order p_0 . $\hat{p}^{\text{HQC}}$ lies above $\hat{p}^{\text{BIC}}$ and increases more apparently with T ; yet it also does not come close to converging to the true lag order. On the other hand, $\hat{p}^{\text{AIC}}$ is consistently much closer to the true lag order and has already nearly reached p_0 at $T = 600$. Therefore, although AIC is not asymptotically consistent, it is the only IC that can come close to the true lag order for realistic sample sizes.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator	AIC	BIC	HQC
	T			
4	240	2.99	1.95	2.06
	360	3.39	2.00	2.16
	480	3.61	2.01	2.40
	600	3.80	2.02	2.66
10	240	6.54	1.96	2.07
	360	8.53	2.00	2.23
	480	8.96	2.00	2.66
	600	9.16	2.02	4.03

Lütkepohl (1985) compares information criteria based on their ability to recover the true lag order. Ivanov and Kilian (2005), by contrast, object to this approach on the

⁶The weights coincide at $N = e^e \approx 15.2$ (for $2 = 2 \ln \ln N$) and never cross for $\ln N$ versus $2 \ln \ln N$, since $x - 2 \ln x > 0$ for all $x > 0$. The condition $\ln N \geq 2$ (BIC versus AIC) holds for $N \geq 8$. Therefore, $N \geq 16$ is the requirement for (14) to hold.

grounds of the bias-variance trade-off discussed above. To see why, it is helpful to consider the decomposition of the mean squared error (MSE) of the estimated median target IRF $\hat{\theta}$ compared to the true DGP IRF θ :

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2] \quad (15)$$

which can be reformulated to:

$$MSE(\hat{\theta}) = E \left[\left(\hat{\theta} - E[\hat{\theta}] \right)^2 + 2 \underbrace{\left(\hat{\theta} - E[\hat{\theta}] \right) \left(E[\hat{\theta}] - \theta \right)}_{=0} + \left(E[\hat{\theta}] - \theta \right)^2 \right] \quad (16)$$

and finally, after we distribute the expected value operator:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2 \quad (17)$$

we have derived an equation for the bias-variance tradeoff. In small samples, if underfitting produces a reduction in variance large enough to offset the resulting increase in bias, the MSE falls, and if we take the MSE as a proxy for accuracy, this matters more than a criterion's ability to recover the true lag order. The lag order selected by a criterion that underfits may therefore outperform the true lag order; one of the central remarks of Ivanov and Kilian (2005).

2.4 Bayesian VAR (BVAR)

So far, we have discussed the theoretical elements necessary to replicate Ivanov and Kilian (2005). We now extend this framework by introducing Bayesian Vector Autoregression. The frequentist lag-order selection problem and Bayesian shrinkage differ fundamentally. The frequentist approach suffers from the curse of dimensionality: each additional lag adds a full $(K \times K)$ block of coefficients to the system, and because estimation must fit these to the data, every estimated parameter consumes one degree of freedom. Frequentist methods therefore demand large samples—a requirement that becomes acute in our smallest design, where a twelve-lag system requires estimating $K^2p = 192$ autoregressive coefficients against only 240 effective observations per equation.

The Bayesian approach reframes this problem. The estimated value of each parameter is determined jointly by the prior and the data, through the posterior. The posterior mean can be understood as a weighted average of the data and the prior,

$$\text{posterior} \approx w(\text{data estimate}) + (1 - w)(\text{prior mean}) \quad (18)$$

where the weight w reflects the tightness of the prior. Because part of each coefficient is

supplied by the prior rather than estimated freely from the data, no parameter consumes a full degree of freedom. The prior does not remove parameters; it constrains them, reducing the effective number of parameters the data must support while leaving the nominal dimension unchanged. The curse of dimensionality is thus addressed not by discarding lags, as selection does, but by disciplining their coefficients—at the cost of bias introduced by pulling each coefficient toward the prior mean.

We follow Lütkepohl (2005, chapt. 5) in introducing the Bayesian VAR methodology. By Bayes’ theorem:

$$g(\boldsymbol{\alpha}|\mathbf{y}) = \frac{f(\mathbf{y}|\boldsymbol{\alpha})g(\boldsymbol{\alpha})}{f(\mathbf{y})} \quad (19)$$

Here, $\boldsymbol{\alpha} := \text{vec}(A_1, A_2, \dots, A_p)$; $g(\boldsymbol{\alpha}|\mathbf{y})$ is the posterior distribution—the distribution of the parameters of interest conditional on the data. The term $f(\mathbf{y}|\boldsymbol{\alpha})$ is the probability density of the sample conditional on the parameters, and is algebraically identical to the likelihood function $l(\boldsymbol{\alpha}|\mathbf{y})$; the two differ only in interpretation, depending on whether $\mathbf{y}$ or $\boldsymbol{\alpha}$ is held fixed. The term $g(\boldsymbol{\alpha})$ is the unconditional distribution of the parameters; since the true distribution is unknown, this component represents our prior belief about the parameters. The denominator $f(\mathbf{y})$ is the unconditional density of the sample; for a given sample, it is simply a normalizing constant. It follows from (19) that the posterior is proportional to the product of the likelihood and the prior:

$$g(\boldsymbol{\alpha}|\mathbf{y}) \propto f(\mathbf{y}|\boldsymbol{\alpha})g(\boldsymbol{\alpha}) = l(\boldsymbol{\alpha}|\mathbf{y})g(\boldsymbol{\alpha}) \quad (20)$$

The practical value of (20) depends on the forms of the two densities on the right-hand side. We assume that the VAR is Gaussian, so that the reduced-form errors are normally distributed ($e_t \sim \mathcal{N}(0, \Sigma)$). Because the process admits a moving-average representation, each y_t is a linear combination of these innovations and is therefore itself normally distributed; the likelihood is consequently Gaussian in $\boldsymbol{\alpha}$. Therefore, the likelihood function may be written as:

$$l(\boldsymbol{\alpha}, \Sigma|\mathbf{y}) = (2\pi)^{-KN/2} |\Sigma|^{-N/2} \exp \left[-\frac{1}{2} \sum_{t=1}^N e_t' \Sigma^{-1} e_t \right], \quad e_t = y_t - v - \sum_{j=1}^p A_j y_{t-j} \quad (21)$$

The only remaining element that determines the posterior is the prior $g(\boldsymbol{\alpha})$. Before specifying it, we must articulate our beliefs about the coefficients. First, in the absence of data, we expect each coefficient to fluctuate around a central value, with no reason to favor one direction of deviation over the other; the prior should therefore be symmetric. Second, we require the posterior to be analytically tractable—available in closed form—because the coefficients must be estimated in each of the many Monte Carlo replications, and reliance on numerical methods would be computationally prohibitive. A normal prior—that is, assuming the prior density of $\boldsymbol{\alpha}$ is Gaussian—satisfies both the

representation of our belief and the requirement of analytical tractability. Thus, we set our prior as:

$$g(\boldsymbol{\alpha}) = \left(\frac{1}{2\pi}\right)^{K^2 p/2} |V_{\boldsymbol{\alpha}}|^{-1/2} \exp \left[-\frac{1}{2}(\boldsymbol{\alpha} - \boldsymbol{\alpha}^*)' V_{\boldsymbol{\alpha}}^{-1} (\boldsymbol{\alpha} - \boldsymbol{\alpha}^*) \right] \quad , \text{ where } \boldsymbol{\alpha} \sim \mathcal{N}(\boldsymbol{\alpha}^*, V_{\boldsymbol{\alpha}}) \quad (22)$$

Since the remaining derivation involves several steps that are not essential to the intuition behind the method, we proceed directly to the result. Product of (21) and (22) yields a normally distributed posterior $\boldsymbol{\alpha} \mid \mathbf{y} \sim \mathcal{N}(\bar{\boldsymbol{\alpha}}, \bar{\Sigma}_{\boldsymbol{\alpha}})$ where the explicit forms of the moments are given by Lütkepohl (2005, chapt. 5):

$$\begin{aligned} w &:= \begin{bmatrix} V_{\boldsymbol{\alpha}}^{-1/2} \boldsymbol{\alpha}^* \\ (I_N \otimes \Sigma^{-1/2}) \mathbf{y} \end{bmatrix}, \\ W &:= \begin{bmatrix} V_{\boldsymbol{\alpha}}^{-1/2} \\ X' \otimes \Sigma^{-1/2} \end{bmatrix}, \\ \bar{\Sigma}_{\boldsymbol{\alpha}} &:= (W'W)^{-1} = [V_{\boldsymbol{\alpha}}^{-1} + (XX' \otimes \Sigma^{-1})]^{-1}, \\ \bar{\boldsymbol{\alpha}} &:= (W'W)^{-1} W'w = [V_{\boldsymbol{\alpha}}^{-1} + (XX' \otimes \Sigma^{-1})]^{-1} [V_{\boldsymbol{\alpha}}^{-1} \boldsymbol{\alpha}^* + (X \otimes \Sigma^{-1}) \mathbf{y}] \end{aligned} \quad (23)$$

Here, $\bar{\boldsymbol{\alpha}}$ is the posterior mean, which we take as our Bayesian VAR coefficient estimate. A remaining difficulty is that computing the posterior mean requires an estimate of the covariance matrix Σ , and a full least-squares estimate of Σ would itself be subject to the curse of dimensionality. Two approaches address this, which we discuss below.

Minnesota Prior

The Minnesota prior is a special case of the normal prior ($\boldsymbol{\alpha} \sim \mathcal{N}(\boldsymbol{\alpha}, V_{\boldsymbol{\alpha}})$) (Doan et al., 1984; Litterman, 1986). We discussed above that the coefficients should fluctuate around the central value. Minnesota prior specifies the central value ($\boldsymbol{\alpha}$) and the amplitude of the fluctuation ($V_{\boldsymbol{\alpha}}$) based on the a priori belief that the macro variables tend to follow a random walk. We use first-differenced variables in our study. Therefore, we follow the Minnesota Prior and form a belief that suggests that our variables follow a white noise procedure (if a series follows a random walk in levels, its first difference is white noise). This implies that we set $\boldsymbol{\alpha}^* = 0$ and $V_{\boldsymbol{\alpha}} \neq 0$. However, the exact specification of $V_{\boldsymbol{\alpha}}$ is itself a substantial topic in the Bayesian VAR literature.

The Minnesota prior encodes its beliefs about the coefficients through the prior variance ($V_{\boldsymbol{\alpha}}$), which is governed by a small set of hyperparameters (See Dieppe et al. (2016) for detailed hyperparameter choices and specifications regarding other priors). Each controls a distinct aspect of how strongly the coefficients are shrunk toward the prior mean. Following Litterman (1986), the prior variance of a coefficient depends on whether it relates a variable to its own lags or to the lags of other variables. We now present the

specification that the Minnesota prior imposes on (V_α) :

$$\sigma_{a_{ii}}^2 = \left(\frac{\lambda_1}{l^{\lambda_3}} \right)^2 \quad (24)$$

Here, $\sigma_{a_{ii}}^2$ is the prior variance for a coefficient relating variable i to lag l (so, for instance, the coefficient of one lagged oil price in the oil price model). λ_1 is the overall tightness hyperparameter—the central choice left to the researcher—which sets the overall strength of the shrinkage. High λ_1 increases the variance (uncertainty around the prior belief), therefore lets the data dominate, and small λ_1 shrinks the variance and prioritizes the prior belief. Two extreme cases are $\lambda_1 = 0$ and $\lambda_1 = \infty$, where the former returns the prior mean (our belief), and the latter returns the OLS estimate (only data matters). l stands for the lag difference (so if it is the prior variance of the coefficient of one lagged oil price in the oil model, $l = 1$) and λ_3 is the lag decay. Lag decay governs how quickly the prior variance declines with the lag length l , through the factor l^{λ_3} . It encodes the belief that coefficients on more distant lags are more likely to be near zero; a larger λ_3 shrinks distant lags more aggressively.

Minnesota prior also specifies the prior variance for a coefficient relating a cross-variable j to lag l :

$$\sigma_{a_{ij}}^2 = \left(\frac{\sigma_i}{\sigma_j} \right)^2 \left(\frac{\lambda_1 \lambda_2}{l^{\lambda_3}} \right)^2 \quad (25)$$

Parameters described above remain the same. There is an additional λ_2 hyperparameter which accounts for cross-variable tightness. This factor, with $0 < \lambda_2 < 1$, applies only to coefficients on the lags of other variables. It encodes the belief that a variable's own past is more informative about its current value than the past of other variables, and therefore shrinks cross-variable coefficients more strongly than its own lags. Setting $\lambda_2 = 1$ removes this distinction, treating own and cross-variable lags symmetrically. $\left(\frac{\sigma_i}{\sigma_j} \right)^2$ component stands for scale adjustment. Because the variables differ in their units and volatility, the ratio of residual standard deviations from univariate autoregressions rescales the cross-variable variances, so that the prior is not distorted by the arbitrary scale of each variable (specifically, σ_i and σ_j are the residual standard deviations of the univariate AR model of each endogenous variable within the system).

We also have deterministic variables (i.e., seasonal dummies and constants) in our equations within the VAR system. Minnesota prior uses the following specification for the prior variances of the coefficients related to deterministic variables:

$$\sigma_{c_i}^2 = \sigma_i^2 (\lambda_1 \lambda_4)^2 \quad (26)$$

In specification (25), the hyperparameter λ_4 sets the prior tightness on the exogenous and deterministic terms—in our specification, the intercept and the eleven monthly seasonal

dummies. These are typically left loosely restricted, reflecting the absence of strong prior beliefs about their values.

The package that we use, **Alexandria**, sets $\lambda_3 = 1$ and $\lambda_4 = 100$, typical values that are used in literature (see Dieppe et al. (2016) and Legrand (2021)). Minnesota prior then still leaves two parameters, λ_1 and λ_2 , to be chosen by the practitioner. We try different λ_1 values: we try fixed parameter values $\lambda_1 \in \{0.2, 0.4, 0.6, 0.8\}$ as well as selecting the λ_1 value from this set that maximizes the Out-of-Sample performance by MDD calculation. The prior that we use (Normal-Wishart Prior, which is discussed below) forces $\lambda_2 = 1$, implying no additional tightness for cross variables.

To sum up, the prior variance structure specified by the Minnesota prior looks like:

$$V_{\alpha}^{\text{Minn}} = \underbrace{\begin{pmatrix} \lambda_1^2 & 0 & \cdots & 0 & \cdots \\ 0 & \left(\frac{\sigma_1}{\sigma_2}\right)^2 (\lambda_1 \lambda_2)^2 & \cdots & 0 & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \left(\frac{\lambda_1}{l^{\lambda_3}}\right)^2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}}_{((K^2P+12K) \times (K^2P+12K)) = ((16P+48) \times (16P+48))} \quad (27)$$

Here, (27) demonstrates prior beliefs regarding variance information of coefficients within the system. We have K^2p lagged variables and 11 seasonal dummies + 1 constant (12 deterministic variables per model). Therefore, we estimate $K^2p + 12K$ coefficients in total.

Under the Minnesota prior with $\alpha^* = 0$, the posterior mean in (23) simplifies to:

$$\bar{\alpha} = [(V_{\alpha}^{\text{Minn}})^{-1} + (X'X \otimes \Sigma^{-1})]^{-1} (X \otimes \Sigma^{-1})\mathbf{y} \quad (28)$$

where $(V_{\alpha}^{\text{Minn}})$ is the Minnesota prior covariance defined in (27). This estimator resembles the least-squares estimator augmented by the prior precision term $(V_{\alpha}^{\text{Minn}})^{-1}$, which shrinks the coefficients toward zero; the strength of this shrinkage is governed by the tightness hyperparameter λ_1 .

The covariance Σ that appears in the posterior mean formula is not taken from the full VAR least-squares estimate - which would itself be subject to the curse of dimensionality - but is obtained from the univariate autoregressions used to set the prior scale (Karlsson, 2012). In our Normal-Wishart implementation, Σ is instead treated as unknown and estimated jointly with the coefficients, as described below.

Normal-Wishart Prior

The Minnesota prior developed above treats the residual covariance matrix Σ as fixed, replacing it with an estimate obtained from univariate autoregressions. This is a sim-

plication: in reality, Σ is unknown and is estimated alongside the coefficients. The Normal-Wishart prior relaxes this assumption by treating Σ as a random parameter and assigning it a prior, so that the uncertainty about Σ is propagated into the posterior rather than ignored. The prior is specified jointly over the coefficients $\boldsymbol{\alpha}$ and the covariance Σ as Dieppe et al. (2016):

$$\Sigma \sim \mathcal{IW}(S_0, \alpha_0), \quad \boldsymbol{\alpha} \mid \Sigma \sim \mathcal{N}(\boldsymbol{\alpha}^*, \Sigma \otimes \Phi_0) \quad (29)$$

The Inverse-Wishart distribution is used to model the residual covariance matrix because a normal distribution cannot guarantee the properties a covariance matrix requires: symmetry and positive definiteness.

In prior (29), Σ is given an Inverse-Wishart prior with scale matrix S_0 and degrees of freedom α_0 . The scale matrix sets the prior mean of the covariance through $E[\Sigma] = S_0/(\alpha_0 - K - 1)$, while α_0 controls the strength of the prior; following Dieppe et al. (2016) we set $\alpha_0 = K + 2$ (here 6), the smallest value yielding a well-defined prior mean. The coefficients ($\boldsymbol{\alpha}$), conditional on Σ , retain a normal prior with mean $\boldsymbol{\alpha}^*$ (set to zero, as before) and covariance $(\Sigma \otimes \Phi_0)$. The matrix $\Phi_0 = \text{diag}\left(\frac{1}{\sigma_1^2} \lambda_1^2, \dots, \frac{1}{\sigma_j^2} \left(\frac{\lambda_1}{\lambda_3}\right)^2, \dots, (\lambda_1 \lambda_4)^2\right)$ is the analogue of V_{α}^{Minn} , but with a crucial difference: it specifies the prior variance for the coefficients of a single equation, which is then scaled across equations by the Kronecker product with Σ (Dieppe et al., 2016).

This conjugate structure, an Inverse-Wishart prior on Σ combined with a conditionally normal prior on $\boldsymbol{\alpha}$, keeps the posterior in the same Normal-Inverse-Wishart family, so that the posterior mean remains available in closed form. The cost of this tractability is a restriction on the prior variance structure, which we now make explicit.

Coefficients' conditioned variance covariance matrix $(\Sigma \otimes \Phi_0)$ has to be modeled in that way in order to assure the existence of the closed form solution (conjugate posterior). Since Φ_0 is the same diagonal matrix for every model, this prior forces the practitioner to set λ_2 hyperparameter equal to one, so that imposing looser or tighter shrinkage on cross-variable coefficients is not possible under the Normal-Wishart prior—the cost of its analytical convenience.⁶

Overall, we use a method in our study that takes the uncertainty of the variance-covariance matrix of the reduced form innovations by enlarging the prior's definition. This approach is more realistic, but it comes at a cost: to preserve conjugacy, the prior covariance of the coefficients must take a specific Kronecker structure. Because Φ_0 is shared across all equations, a given position in Φ_0 corresponds to an own-lag coefficient in one equation but a cross-variable coefficient in another; the single shared Φ_0 therefore cannot assign them different shrinkage, which forces $\lambda_2 = 1$. That is why Dieppe et al. (2016) suggests that the λ_1 hyperparameter should be set more tightly when the Normal-Wishart prior is used.

Having established the theoretical framework and estimation methodology, we now turn to the simulation design to evaluate the empirical performance of the competing approaches.

3 Simulation Design

This Monte Carlo simulation study follows Ivanov and Kilian (2005) design with several alterations in order to make our extensions feasible and to fit the needs of our analysis. First of all, we only use one monthly DGP based on the empirical VAR study of Kilian and Murphy (2014). Therefore, we do not get estimations for quarterly VAR series or VEC series, and additionally, we do not have additional robustness by including several different series with different macroeconomic indicators and movements. The reason for this limitation is our tight time constraint. We use the dataset from Kilian and Murphy (2014) containing the following 4 variables: percent change in global oil production, real activity index from Kilian (2009), the log real price of oil, and changes in OECD crude oil inventories. Just as in the original study, we first estimate a VAR with a set lag order of these 4 variables and then use the sign restrictions approach explained earlier to identify structural IRFs and select the median target.

The DGP is formed by structural VAR model using the structural impact matrix $\tilde{B}$ that corresponds to the median target IRF as the parameter for the structural shocks e_t , which are drawn from an i.i.d. standard normal distribution. The formula for the DGP is represented by a structural VAR model using the estimated reduced form parameters A (for the lag coefficients y_{t-p}) and V (for the monthly dummy parameters d_t) and the structural impact matrix $\tilde{B}$ as follows:

$$y_t = Vd_t + Ay_{t-p} + \tilde{B}e_t \quad (30)$$

We estimate these parameters using the original data for two different lag orders, $p_0 \in \{4, 10\}$ and create DGPs with different sample sizes, $T \in \{240, 360, 480, 600\}$, where $T = N - \bar{p}$. $\bar{p}$ is the set of lag orders that are considered for both the IC and the BVAR estimations; it is uniformly set to $\bar{p} = 12$ to ensure that the true lag order p_0 is part of the set. We restrict the lag order selection p_0 to only include one high and one low lag order and drop $T = 300$ in comparison to the setup of Ivanov and Kilian (2005) in order to save computational power for the sign restriction decomposition. For sampling, we use random draws of length p_0 from the original dataset as initial values, then apply the parametric bootstrap using the SVAR described above and generate sequences of lengths T .

On each of these samples, VARs fitted by lag orders selected by both the three ICs as well as different specifications of BVAR are estimated, a fixed number of admissible IRFs

found, and the median target IRF determined. We estimate all BVARs using $p = \bar{p} = 12$ as the lag order, thereby solely relying on shrinkage for reducing the variance and bias of the estimation.

Finally, we estimate the key performance parameter, the relative MSE, by calculating the geometric mean over the ratio of the squared errors of the median target IRFs $\hat{\theta}_j$ computed by method j in relation to the squared errors of the median target IRF by a model fit with the true lag order $\hat{\theta}_{p_0i}$ over all Monte Carlo iterations from 1, ..., R :

$$MSE_j^{relative} = \exp \left(\frac{1}{n} \sum_{i=1}^R \ln \left(\frac{SE_{ji}}{SE_{p_0i}} \right) \right) \quad (31)$$

The geometric mean is preferred over the arithmetic mean as the ratios are bounded on the left side at 0, but unbounded on the right side. Outliers on the right side are hence overweighed in the arithmetic mean. Normalization by the OLS VAR estimation with p_0 ensures that the performance in each iteration is weighed by how well a correctly identified model performs.

For the computation of the DGP parameters, we set the number of selected admissible IRFs to 1 million in order to get a highly accurate representation of the real world movements. However, during the Monte Carlo simulation we have to reduce that number to 1,000, which is seen as the minimum requirement for accurate set identification, albeit still with a larger variance. We set the number of Monte Carlo iterations to $R = 500$. We understand the concern that this may be too little for eliminating all Monte Carlo biases. Yet we have to rely on such a small number as it is the absolute maximum that our hardware allows for, even though we have optimized the code by employing parallel processing and delegation to C-based processes using the `numba` and `joblib` packages in `python`. All in all, this simulation still took 18 hours to complete with 6.5 million admissible IRFs drawn.

4 Results

First, we will look at the main comparison of performance between the different ICs and BVAR specifications in terms of recovering an accurate median target IRF. Table 2 displays the full set of relative MSEs. It reveals two main findings:

1. We confirm the results from Ivanov and Kilian (2005) about the performance of the different information criteria. AIC consistently outperforms both BIC and HQC, revealing that the strict penalization of the latter two towards low lag orders is biasing the IRF too much. On the other hand, AIC's milder lag penalization ensures a closer fit to the true lag order p_0 for finite samples, thereby reducing bias. The reduction of the variance by selecting a p much lower than the true lag order as done by BIC and HQC, see Table 1, is not evidenced to reduce the coefficient variance enough to compensate for

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage.

p_0	T	AIC	BIC	HQC	BVAR ($\lambda_1 = 0.2$)	BVAR ($\lambda_1 = 0.4$)	BVAR ($\lambda_1 = 0.6$)	BVAR ($\lambda_1 = 0.8$)	BVAR (MDD λ_1)
4	240	1.083	1.108	1.129	1.062	1.055	1.076	1.145	1.062
	360	1.117	1.274	1.296	1.030	1.167	0.994	1.142	1.030
	480	0.986	1.133	1.124	1.137	1.109	1.092	1.100	1.137
10	600	0.989	1.145	1.118	1.133	0.983	1.103	1.109	1.133
	240	1.033	1.018	1.021	0.895	0.887	0.868	0.947	0.895
	360	0.811	0.904	0.878	0.896	0.900	0.856	0.905	0.912
	480	0.989	1.190	1.185	0.853	0.920	0.876	0.944	0.866
	600	1.036	1.260	1.259	0.810	0.895	0.848	0.819	0.825

Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.

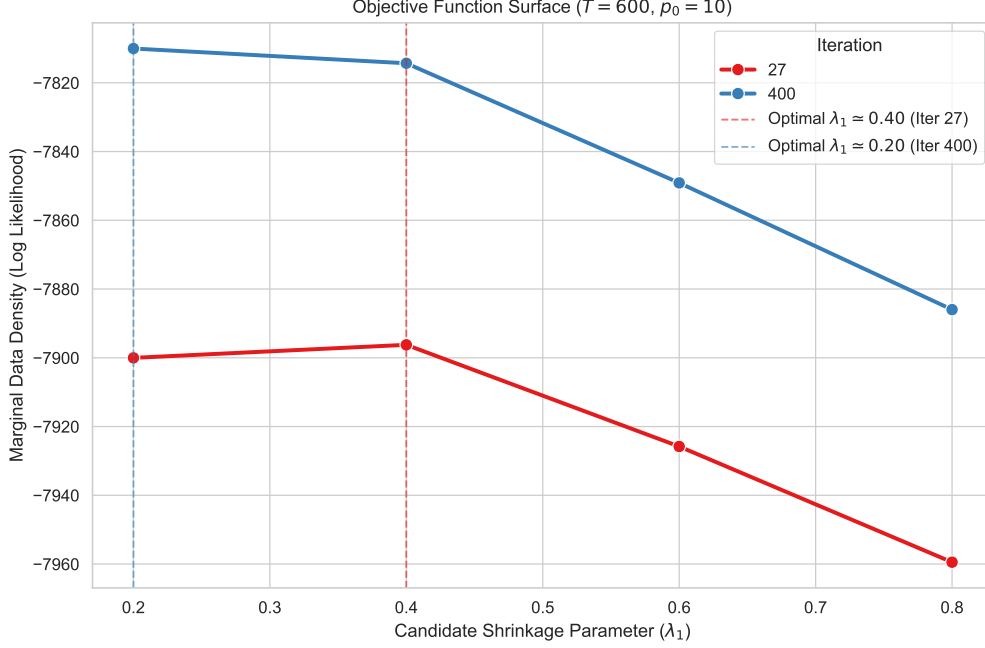


Figure 1: Objective Function Surface for Marginal Data Density Optimization for two randomly selected iterations at $p_0 = 10$ and $T = 10$.

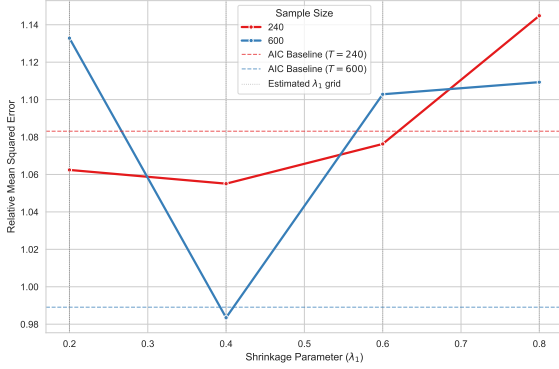
the additional bias.

2. BVAR dominates the three classical lag selection criteria across most scenarios; only in two cases, AIC remains the leader (see Table 2). For a better comparison among the different BVAR specifications, Table 3 and Figure 2 present the results in terms of relative MSE compared to the overall best performing information criterion, AIC. It reveals that BVARs have a stronger advantage over AIC if the true lag order p_0 is large. Except for an outlier for $T = 360$, BVAR can reduce the MSE strongly across all different specifications for λ_1 in the case of $p_0 = 10$. A better λ_1 selection can still improve the estimation; yet it is not crucial to defend the usage of BVAR over AIC. On the other hand, for the low lag order of p_0 , the benefit of BVAR is not as clear. Here, whether it is recommended to choose BVAR over classical OLS estimation with AIC depends on the hyperparameter selection λ_1 . For example, while the selection of $\lambda_1 = 0.4$ leads to a slight improvement of 0.6% at $T = 600$, other selections lead to a loss of accuracy of more than 10%. These findings show that an OLS model with AIC-selected lag order can still work fairly well for a low lag order p_0 in finite samples, while the disadvantage of this estimation is strong for high p_0 . This conclusion is consistent with the bias-variance tradeoff presented before. $\hat{p}^{AIC}$ only slightly underfits the model in finite samples of this magnitude. For a low lag order, OLS can still estimate all parameters with relatively small variance. However, for a high lag order such as $p_0 = 10$, estimating all parameters produces unstable estimates. Here, the shrinkage of further lag orders introduced in BVAR is a clear advantage. It enables the model to capture the signal from higher lag

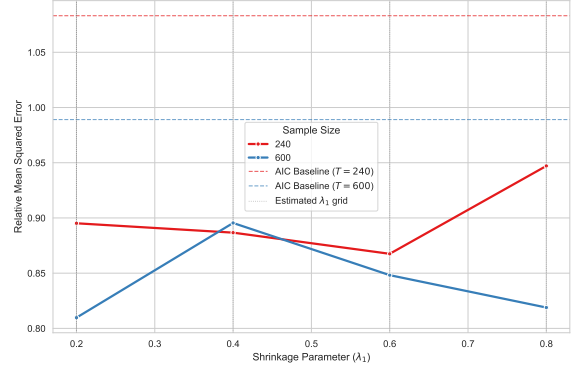
Table 3: Relative Mean Squared Error of Bayesian Shrinkage Specifications vs. AIC Baseline.

p_0	T	BVAR ($\lambda_1 = 0.20$)	BVAR ($\lambda_1 = 0.40$)	BVAR ($\lambda_1 = 0.60$)	BVAR ($\lambda_1 = 0.80$)	BVAR (MDD λ_1)
4	240	0.981	0.974	0.994	1.057	0.981
	360	0.922	1.045	0.890	1.022	0.922
	480	1.153	1.125	1.108	1.116	1.153
	600	1.145	0.994	1.115	1.122	1.145
10	240	0.866	0.858	0.840	0.917	0.866
	360	1.105	1.110	1.055	1.116	1.124
	480	0.863	0.931	0.886	0.955	0.876
	600	0.782	0.865	0.819	0.791	0.796

Note: Bold values indicate the lowest Relative Mean Squared Error across the respective scenario.



(a) $p_0 = 4$



(b) $p_0 = 10$

Figure 2: Comparison of BVAR's performance across different λ_1 -values in comparison to AIC.

orders without adding too much noise. Additionally, as we follow the common approach to fit a general high lag order of $\bar{p} = 12$ to both cases of p_0 , the BVAR for $p_0 = 4$ includes more lags that are beyond the true underlying limit. Although they are shrunk towards 0 based on the prior and the computed posterior probabilities, they still introduce a bias.

Over both p_0 s, we find that the BVAR estimation using $\lambda_1 \in \{0.2, 0.4, 0.6\}$ has more instances of outperforming than underperforming AIC, while $\lambda_1 = 0.8$ loses the comparison to the AIC benchmark in the majority of cases.

Overall, the medium-loose prior $\lambda_1 = 0.6$ is the best hyperparameter in four cases. Both $\lambda_1 = 0.2$ and $\lambda_1 = 0.4$ are dominant in two scenarios. The very loose prior of $\lambda_1 = 0.8$ is shown to be optimal in no case. There is no absolutely clear pattern in terms of sample size or true lag order for why certain λ_1 -values perform better in certain scenarios than in others, neither when you consider all scenarios nor when you separate the analysis by p_0 .

This may be explained by two reasons. First, the inherent bias-variance trade-off in shrinkage implies that both strong shrinkage, through reduction of variance, as well as weak shrinkage, through reduction of bias, can make the estimation more precise. This may explain results in certain scenarios, such as the combination of p_0 and $T = 10$, where the relative MSE of the tightest hyperparameter, $\lambda_1 = 0.2$, is very close to that of the loosest, $\lambda_1 = 0.8$. $\lambda_1 = 0.8$ ensures that higher lag orders are still more relevant for the estimation, making it fit better to the true model of lag order $p_0 = 10$. Meanwhile, $\lambda_1 = 0.2$ can reduce the variance of low lag orders, making a few coefficients more accurate while neglecting others. Second, the hardware-forced limitations on the number of iterations and the number of required admissible IRFs per model may make the results too unstable. It can therefore not be refuted that some of the variation among our results is driven by random sampling variations.

Interestingly, the MDD-optimized λ_1 also never takes first place among the BVAR models. Figure 1 reveals the reason. For low sample sizes T as well as the lower true $p_0 = 4$, MDD purely selects the lowest option $\lambda_1 = 0.2$, which has been shown not to perform the best in these cases. Only for $p_0 = 10$ does it select a higher $\lambda_1 = 0.4$ in some iterations. However, both $\lambda_1 = 0.2$ and $\lambda_1 = 0.4$ do not dominate for these scenarios. Again, the weak performance of these specific λ -values might be partially attributed to sampling error. Nevertheless, it shows that MDD might not necessarily select the optimal hyperparameter for the purpose of the inference of structural relationships by VAR, as its optimization on out-of-sample performance generally leads to a preference towards stronger shrinkage, which might not be optimal for constructing IRFs, as it eliminates the signal from higher lag orders.

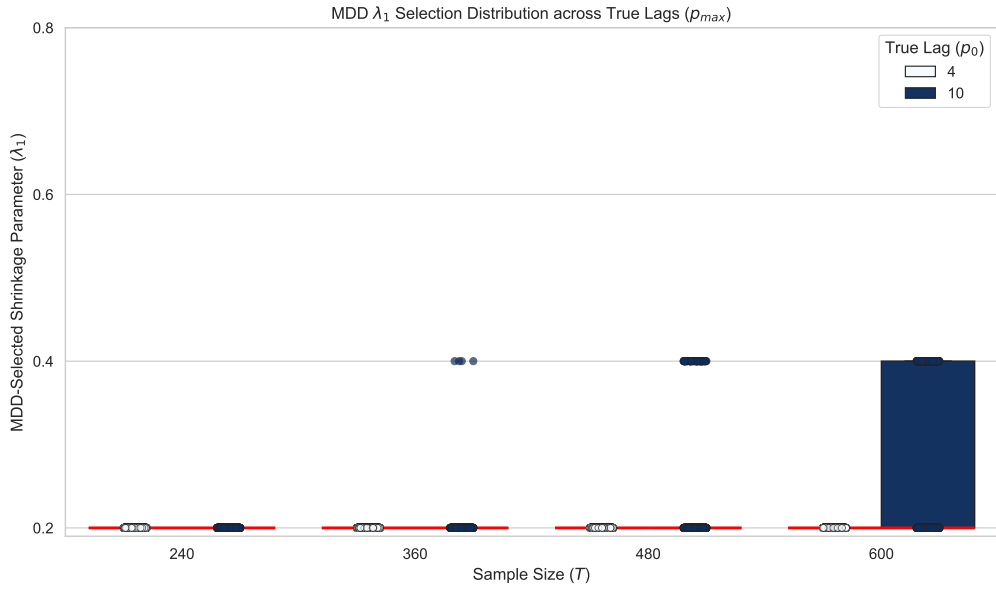


Figure 3: The MDD selected priors for different sample sizes and both $p_0 = 4$ and $p_0 = 10$.

Note: Circles represent optimal λ_1 -values in individual iterations, bars represent the interquartile range, red lines indicate the median.

5 Conclusion

6 Conclusion

This study evaluates the specification of Structural Vector Autoregressions (VARs) for the purpose of impulse response analysis, updating the foundational framework of Ivanov and Kilian (2005) to reflect contemporary empirical practices. By constructing a realistic data-generating process based on the global oil market model of Kilian and Murphy (2014) and employing set identification via sign restrictions, we compare the efficacy of traditional information criteria against Bayesian VAR (BVAR) shrinkage in recovering the true median target Impulse Response Function (IRF).

Our Monte Carlo simulation results confirm that among traditional information criteria, the Akaike Information Criterion (AIC) consistently outperforms the Bayesian (BIC) and Hannan-Quinn (HQC) criteria in finite samples. The stricter penalization of the BIC and HQC leads to severe underfitting, introducing a bias that outweighs the benefits of reduced variance. In contrast, the AIC’s milder penalty allows for a closer approximation of the true lag order p_0 , making it the most robust frequentist choice for structural inference.

More importantly, our findings demonstrate that BVAR models mostly dominate ordinary least squares estimations reliant on IC-selected lag orders. This advantage becomes particularly pronounced when the true underlying lag order is large (e.g., $p_0 = 10$). Rather than strictly truncating the lag length and risking omitted variable bias, the BVAR approach retains higher-order lags while applying prior shrinkage to discipline the parameter space. This continuous bias-variance trade-off consistently yields more accurate structural IRFs than the binary selection mechanism of information criteria.

However, the optimal degree of shrinkage remains ambiguous. We find no strictly dominant hyperparameter λ_1 across all sample sizes and true lag orders, as both tighter and looser priors can successfully minimize the mean squared error through different channels—variance reduction and bias reduction, respectively. While a medium-loose prior ($\lambda_1 = 0.6$) most frequently achieved the lowest relative MSE in our simulations, the performance differentials were sometimes marginal. We suspect that residual sampling variance, driven by computational constraints that limited our Monte Carlo study to 500 iterations, contributes to this ambiguity. Future research leveraging greater computational resources could further clarify these parameter boundaries by enabling a higher volume of iterations and admissible IRF draws.

Finally, we highlight a critical divergence between forecasting and structural analysis objectives. Optimizing the shrinkage parameter λ_1 via Marginal Data Density (MDD)—a metric geared toward out-of-sample forecasting—tends to favor excessively tight priors. While this heavy shrinkage successfully minimizes out-of-sample prediction error,

it overly dampens the signal from higher-order lags, yielding suboptimal estimations of the structural IRFs. Ultimately, we show that specification strategies designed to maximize forecasting accuracy do not necessarily align with the goal of accurately identifying macroeconomic structural relationships. For practitioners prioritizing structural inference over pure forecasting, BVAR estimation leveraging a moderately loose prior provides an effective methodological approach.

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Appendix

A DGP IRFs

Figure 4 visualizes the IRFs selected for the DGP of $p_0 = 4$ and $p_0 = 10$. You can see that they exhibit the same general pattern but that a higher lag order captures more complex movements.

B Derivation of the Relationship Between Structural Form Innovation Terms and Reduced Form Errors

For clarity, several intermediate steps were omitted in the main text. We collect them here. To derive the relationship in equation (2), we begin from the structural form:

$$Y_t = B_0 Y_t + B_1 Y_{t-1} + B_2 Y_{t-2} + \cdots + B_{p_0} Y_{t-p_0} + \varepsilon_t \quad (32)$$

Moving the contemporaneous term $B_0 Y_t$ to the left-hand side gives:

$$\underbrace{(I - B_0)}_{B^*} Y_t = B_1 Y_{t-1} + B_2 Y_{t-2} + \cdots + B_{p_0} Y_{t-p_0} + \varepsilon_t \quad (33)$$

Premultiplying by the inverse of B^* yields the reduced form:

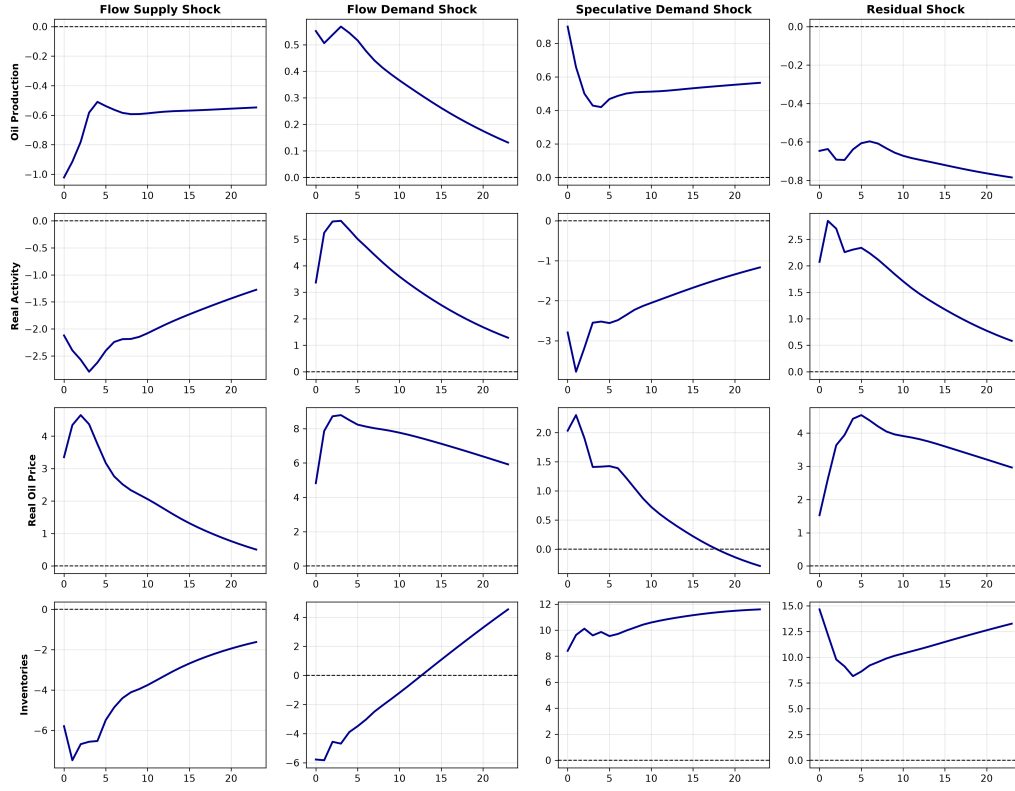
$$Y_t = \underbrace{(B^*)^{-1} B_1}_{A_1} Y_{t-1} + \underbrace{(B^*)^{-1} B_2}_{A_2} Y_{t-2} + \cdots + \underbrace{(B^*)^{-1} B_{p_0}}_{A_{p_0}} Y_{t-p_0} + \underbrace{(B^*)^{-1} \varepsilon_t}_{e_t} \quad (34)$$

which is the explicit reduced-form representation. The relationship defined in (2) is now apparent: writing $(B^*)^{-1} = B$, we have:

$$e_t = B \varepsilon_t \quad (35)$$

Here, B_0 and B denote distinct objects: B_0 is the structural contemporaneous coefficient matrix, which governs the simultaneous interactions among the endogenous variables in the structural form, whereas $B = (I - B_0)^{-1}$ is the impact matrix that maps the unit-variance structural shocks ε_t into the reduced-form innovations e_t . The former is a primitive of the structural model; the latter is the object recovered through the identification procedure in the main text.

True Structural Impulse Responses (4-Lag Model, Median Target Model)



True Structural Impulse Responses (10-Lag Model, Median Target Model)

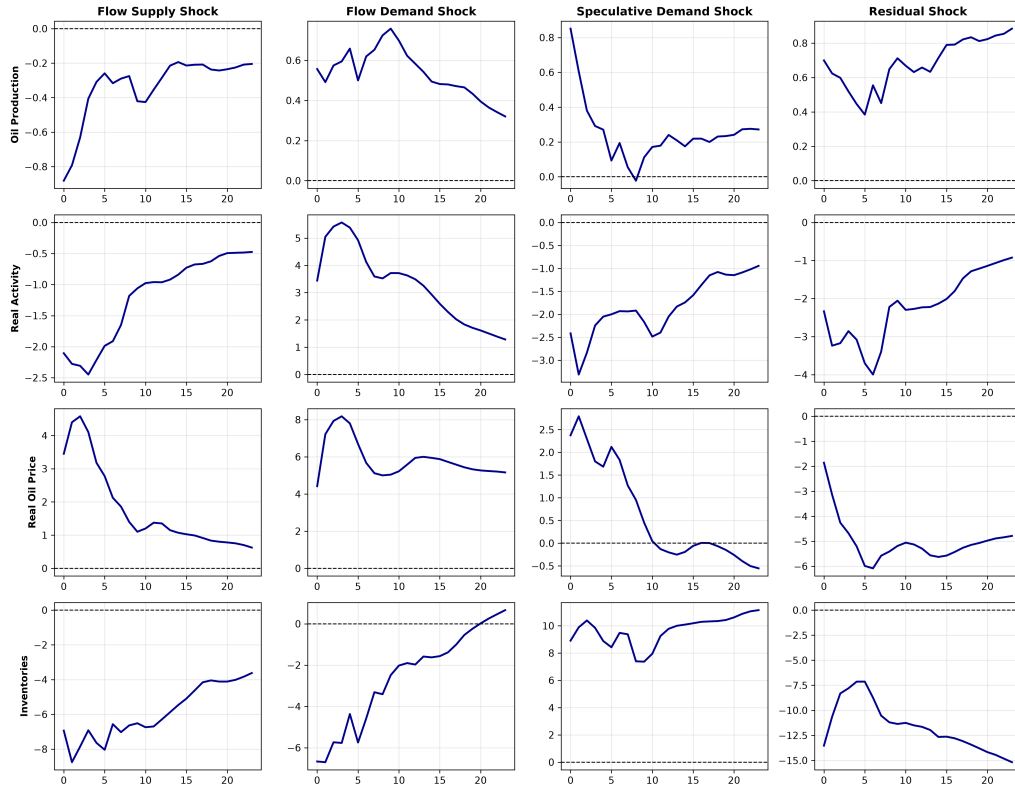


Figure 4: DGP IRFs for $p_0 = 4$ and $p_0 = 10$

C Derivation of the Recursive Coefficients and the Mean

We now derive the recursion in (4). It is convenient to define the autoregressive lag polynomial:

$$A(L) = I - A_1L - A_2L^2 - \dots - A_{p_0}L^{p_0} \quad (36)$$

Its inverse admits a power-series representation in the lag operator:

$$A(L)^{-1} = \Phi(L) = \sum_{i=0}^{\infty} \Phi_i L^i = \Phi_0 + \Phi_1 L + \Phi_2 L^2 + \dots \quad (37)$$

Which leads to:

$$A(L)A(L)^{-1} = A(L)\Phi(L) = (I - A_1L - A_2L^2 - \dots)(\Phi_0 + \Phi_1 L + \Phi_2 L^2 + \dots) = I \quad (38)$$

Matching coefficients on equal powers of L gives:

$$\begin{aligned} I\Phi_0 &= I_K \implies \Phi_0 = I_K \\ \Phi_1 - A_1\Phi_0 &= 0 \implies \Phi_1 = A_1\Phi_0 \\ \Phi_2 - A_1\Phi_1 - A_2\Phi_0 &= 0 \implies \Phi_2 = A_1\Phi_1 + A_2\Phi_0 \end{aligned}$$

and, in general, the recursion:

$$\Phi_i = \sum_{j=1}^i A_j \Phi_{i-j} \quad (39)$$

The unconditional mean is $\mu = (I - A_1 - A_2 - \dots - A_{p_0})^{-1}v$ provided the process is weakly stationary. To derive it, take expectations of (1) under weak stationarity, so that $\mathbb{E}[Y_t] = \mu$ for all t :

$$\mathbb{E}[Y_t] = v + A_1\mathbb{E}[Y_{t-1}] + A_2\mathbb{E}[Y_{t-2}] + \dots + A_{p_0}\mathbb{E}[Y_{t-p_0}] \iff [I - A_1 - A_2 - \dots - A_{p_0}]\mu = v \quad (40)$$

which gives:

$$\mu = [I - A_1 - A_2 - \dots - A_{p_0}]^{-1}v \quad (41)$$

D Detailed Review of Normal-Wishart Prior

The VAR model can be expressed in terms of the Kronecker product:

$$y = (X' \otimes I_K)\alpha + e \quad (42)$$

Where $\alpha = \text{vec}(A)$ that contains all the coefficients and $e \sim N(0, I_N \otimes \Sigma)$. The same Gaussian likelihood can be written using the Kronecker product as:

$$\ell(\alpha, \Sigma | y) \propto |\Sigma|^{-N/2} \exp \left[-\frac{1}{2}(\alpha - \hat{\alpha})'(X'X \otimes \Sigma^{-1})(\alpha - \hat{\alpha}) \right] \exp \left[-\frac{1}{2}\text{tr}(\Sigma^{-1}\hat{S}) \right] \quad (43)$$

Where $\hat{\alpha}$ is the OLS estimate and $\hat{S} = (Y - X\hat{A})'(Y - X\hat{A})$. The key point is that Σ^{-1} and $X'X$ enter jointly. If we set a Minnesota prior as in (22):

$$\pi(\alpha) \propto \exp \left[-\frac{1}{2}(\alpha - \alpha^*)'V_\alpha^{-1}(\alpha - \alpha^*) \right] \quad (44)$$

and multiply (43) by (44) to obtain the posterior, the exponent becomes:

$$(X'X \otimes \Sigma^{-1}) + V_\alpha^{-1} \quad (45)$$

The first term carries Σ^{-1} in a Kronecker structure; the second does not contain Σ at all. This is problematic because Σ^{-1} cannot be factored out. Conjugacy is lost because the coefficient and covariance components are no longer separable: the α terms and Σ terms remain intermingled. If instead we specify $V_\alpha = \Sigma \otimes \Phi_0$, exponent becomes:

$$(X'X \otimes \Sigma^{-1}) + (\Phi_0^{-1} \otimes \Sigma^{-1}) \quad (46)$$

so that Σ^{-1} can be factored out:

$$(X'X + \Phi_0^{-1}) \otimes \Sigma^{-1} \quad (47)$$

The α and Σ components are now separate, and the posterior has a closed-form solution.

It is also important to discuss why is λ_2 inoperative. Consider the prior variance structure of Minnesota Prior:

$$v_{ij,l} = \begin{cases} \left(\frac{\lambda_1}{l^{\lambda_3}}\right)^2 & i = j \text{ (own lag)} \\ \left(\frac{\sigma_i}{\sigma_j}\right)^2 \left(\frac{\lambda_1\lambda_2}{l^{\lambda_3}}\right)^2 & i \neq j \text{ (cross-variable)}. \end{cases} \quad (48)$$

Because the full prior covariance V_α^{Minn} is diagonal with one entry per coefficient, the own and cross-variable cases can be assigned different values: the factor $0 < \lambda_2 < 1$ appears only in the $i \neq j$ entries, shrinking cross-variable coefficients more strongly.

Under $\Sigma \otimes \Phi_0$, the prior variance of a coefficient is no longer assigned per coefficient. Consider the two-variable, one-lag case for concreteness:

$$\begin{aligned} y_{1t} &= \alpha_{11}y_{1t-1} + \alpha_{12}y_{2t-1} + e_{1t} \\ y_{2t} &= \alpha_{21}y_{1t-1} + \alpha_{22}y_{2t-1} + e_{2t} \end{aligned} \quad (49)$$

The single-equation template Φ_0 is diagonal with one entry per regressor position:

$$\Phi_0 = \text{diag}(\phi_1, \phi_2) \quad (50)$$

Where, ϕ_m is the prior variance attached to position m within any equation. The full coefficient covariance is then:

$$\Sigma \otimes \Phi_0 = \begin{pmatrix} \sigma_{11}\Phi_0 & \sigma_{12}\Phi_0 \\ \sigma_{21}\Phi_0 & \sigma_{22}\Phi_0 \end{pmatrix} \quad (51)$$

Using (51), we can calculate the prior variances of coefficients. $\text{var}(\alpha_{11}) = \sigma_{11}\phi_1$, $\text{var}(\alpha_{12}) = \sigma_{11}\phi_2$, $\text{var}(\alpha_{21}) = \sigma_{22}\phi_1$ and $\text{var}(\alpha_{22}) = \sigma_{22}\phi_2$. Each of ϕ_1 and ϕ_2 is shared between an own (autoregressive) lag in one equation and a cross-variable lag in the other: ϕ_1 applies to the own lag α_{11} and the cross lag α_{21} , while ϕ_2 applies to the cross lag α_{12} and the own lag α_{22} . Because a single template entry cannot shrink own and cross lags differently, $\lambda_2 = 1$ is forced.

Declaration of AI Usage

During the preparation of this seminar paper, I, [NAME], used Claude, Gemini, and ChatGPT for help with coding, research, and adapting writing to academic standards. I have reviewed and edited the content as needed and take full responsibility for the content.

[NAME]